

Features/Platform	Ethereum	OpenChain	BigChainDB	HydraChain	Corda	Multichain
Security	Spying data is not possible	Multiple levels of control	Highly secure with transaction level permissions	Extension of Ethereum platform architecture	Higher privacy for data and transaction	Moderate security with privacy features for transaction
Support/Governance	Ethereum Consortium	Hierarchical system to set permission at any level	Federation Consensus Model	Ethereum consortium	Corda Consortium	Global consortium
Community Activity	Democratic Autonomous Organizations (DAO)	Slack community development	Community development in Git	Open community	Slack and StackOverflow	Git based community work
Limitations	Open financial system access	Unified API support and extensibility is not available	Data oriented solution and no industry focus	In presence of pending transaction only a new block gets created	No global broadcast of transaction across network	Flexible to extend privacy features is not supported
Industry Focus	Multi-Industry application	Insurance and Banking	Multi-Industry application	Financial services	Banking, Financial Services, Insurance	Multi-Industry application
Type of Ledger	Permissionless	Permissionless	Permissioned or private network	Permissioned	Permissioned	Permissioned or Private network
Functionality	Digital Asset Management	Digital Asset Management	Big Data management	Customizable fee, genesis allocation, block time	Pluggable consensus	Data Ledger functions
Smart Contract	Ethash - Proof of Work (PoW)	Partitioned Consensus	Federation of Voting nodes	Proof of Work (PoW)	Supported	Proof of Work (PoW)
Crypto Currency used	ETH	No Native currency.	No Native currency.	ETH	No Native currency.	No Native currency.

Figure 8: Comparison of open source blockchain platforms

blockchains, either within or between organisations. It aims to overcome a key obstacle to the deployment of blockchain technology in the institutional financial sector, by providing the privacy and control required in an easy-to-use package.

Key highlights

- **Activity:** Medium but actively updated on GitHub
- **Type of ledger:** Private, permissioned
- **Pricing:** Open source
- **Supported languages:** Python, C#, JavaScript, PHP, Ruby
- **GitHub repo:** savior (Python); c# MultichainLib (C#); Multichain-Node (JavaScript); libphp-multichain (PHP); multichain-client (Ruby)

Functionality

The functional architecture of the blockchain platform is based on various consensus algorithms, which are used as the basis of the platform execution mechanism. These include the following.

Proof of Work (PoW): This is the original type of consensus algorithm from Satoshi Nakamoto's first research on blockchain. This algorithm is used as a mechanism to confirm the transaction completeness and if successful, a new block gets added to the chain of network. This

helps any Distributed Denial of Service (DDoS) network attacks.

Proof of Stake (PoS): Since PoW involves a lot of system resources like GPU, it is an advanced mechanism suitable for sophisticated blockchain networks. PoS solves the challenge of this high resource usage by using an algorithm based on distributed tokens and static coin supply.

Delegated Proof of Stake (DPoS): This is an extended version of PoS. The PoS is like a lottery mechanism based on the coin stake of the users and DPoS is based on influencing the network by all stakeholders equally. It is based on an algorithm approach that balances electing a list of nodes by any stakeholder to complete the transaction.

There are other consensus mechanisms like Practical Byzantine Fault Tolerance (PBFT) and Directed Acyclic Graph (DAC), which are evolving in the Blockchain 3.0 wave (Blockchain 1.0 is Bitcoin architecture and Blockchain 2.0 is Ethereum consensus architecture). They are getting implemented in platforms like HyperLedger and Ripple.

When any transaction happens in a blockchain platform, it gets validated either by permissionless or permissioned blockchain. In permissionless blockchains, the transaction is validated by the public, who are participants in the network and hence the transaction processing rate is higher at any given time. In permissioned blockchain, the transaction is validated by a selected group of participants and approved by the blockchain owner or approver.

The table in Figure 8 compares various features and architectural points of the blockchain platforms like security, support, community activity, limitations in platform support, industry-focus of the solution and the type of ledger used in the platform, as well as the functionality supported. 

References

- [1] <https://ethereum.org/>
- [2] <https://www.openchainproject.org/>
- [3] <https://www.bigchaindb.com/>
- [4] <https://www.hyperchain.cn/en>
- [5] <https://www.corda.net/>
- [6] <https://www.multichain.com/>

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